

Data Mining Project Background Instructions

For this project, you will work within your existing capstone groupings and focus solely on the data mining component of your capstone project.

Your task is to write a background section explaining how data mining techniques are applied within your project and demonstrate your findings through graphs and a simulation video.

Your Background Section Must Include:

1. Project Overview

- Briefly describe your capstone project.
- Define the problem or challenge your project aims to address.
- Identify the industry or area of application.

2. Role of Data Mining

- Explain why data mining is relevant to your project.
- Describe the types of data you will be working with.
- Discuss the potential benefits data mining can provide.

3. Data Mining Techniques Used

- Identify which data mining methods (classification, clustering, association rules, etc.) you will apply.
- Justify your choice—why is this technique best suited for your project?

4. Expected Outcomes, Graphs & Simulation Demonstration

- Perform a simulation showcasing how your chosen data mining technique processes and extracts meaningful insights from your dataset.
- Include graphs illustrating key findings from your simulation (such as patterns, trends, or clusters within the data).
- Record a video demonstration of your simulation, explaining the inputs, processing steps, and how the graphs visualize your results.
- Discuss the insights gained from the simulation, explaining their significance and impact on decision-making within your project.

Formatting Guidelines:

- The document should be clear and well-organized.
- Use headings and subheadings to structure your background.
- Ensure proper grammar and coherence in your writing.
- Cite any sources where applicable.

Submission Details:

- Written Background Document in PDF
- Graphs: Integrated within the document
- Simulation Video
- Due Date: May 29, 2025 (to be submitted via LMS)

Note: You are not required to develop new datasets. Instead, apply data mining techniques to the existing datasets relevant to your capstone project.